Type MS · Identify the independent or dependent variable

Answer 1 of 10 · Question 4 part b.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
(ii)	lipase		1
(iii)	fatty acids; glycerol;		2
(b) (i)	Any value between 8.6 and 8.8;		1
(ii)	independent = time/ temperature; dependent = pH;		2
(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Identify the independent or dependent variable

Answer 1 of 10 · Question 4 part b.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 2 of 10 · Question 8 part b.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

8 (a)(i)	• high (blood) pressure (in the glomerulus); • small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule);	Do not allow protein	1 1
(ii)	Any four from: • lower concentration of glucose/oxygen in blood in renal vein; • used in respiration; • less/no urea in renal vein; • urea excreted in urine; • more carbon dioxide in renal vein; • carbon dioxide produced in respiration;	Allow reverse argument throughout	Max 4
(b) (i)	quantity of salt (in food);		1
(ii)	Any two from: • different foods used (which may affect the amount of urea produced/amount of water reabsorbed); • different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance); • the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced);		Max 2
(iii)	use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content);	Allow valid alternatives	1

Total question 8 = 10 marks

Type MS · Identify the independent or dependent variable

Answer 2 of 10 · Question 8 part b.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 3 of 10 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

4 (a)	(i)	• axes labelled (1) • suitable scale (1) • axes correct way round (1) • correct plots (1) • lines labelled/key given (1)		5
	(ii)	any two from • volume of water (1) • mass of carbohydrate/aspirin (1) • temperature (1) • time interval between each reading (1)	Allow type of aspirin Allow amount as an alternative to mass Ignore concentration	2
	(iii)	pH (1)		1
(b)		• (aspirin) dissolves/reduces pH more quickly (1) • (when combined) with sucrose (1) • so will act more quickly/relieves pain faster (1)		3

Total 11 marks

Type MS · Identify the independent or dependent variable

Answer 3 of 10 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 4 of 10 · Question 7 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · Identify the independent or dependent variable

Answer 4 of 10 · Question 7 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 5 of 10 · Question 6 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

6 (a)	(type of) antibacterial cleaner;		1
(b)	Any two from the following • use sterilised equipment/named equipment; • heat the inoculating loop (in a Bunsen flame); • do not open the lid of the Petri dish too much; • seal the lid of the Petri dish; (with tape)	Named equipment must be hardware	2
(c)	• to prevent growth/reproduction; • of pathogen/harmful microorganism;		2
(d)	it is a control;		1
			Total 6

Type MS · Identify the independent or dependent variable

Answer 5 of 10 · Question 6 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 6 of 10 · Question 5 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

5 (a)	(i)	any five from • place substrate into a test/boiling tube/beakers; • place test tube in a (hot) water bath; • add enzyme to substrate; • measure time taken for reaction to occur; • use same volume/concentration of enzyme/substrate; • repeat for other temperatures;	Allow a temperature value Allow example of a reaction e.g. colour change, amount of glucose produced reject amount	(5)
	(ii)	temperature;		(1)
(b)		• less kinetic energy; • less/fewer collisions (with substrate); • less enzyme-substrate complexes formed/less binding; • lower enzyme activity/rate of reaction;		(4)

Total for question 5 = 10 marks

Type MS · Identify the independent or dependent variable

Answer 7 of 10 · Question 2 part c · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Question number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · Identify the independent or dependent variable

Answer 7 of 10 · Question 2 part c · 1 mark

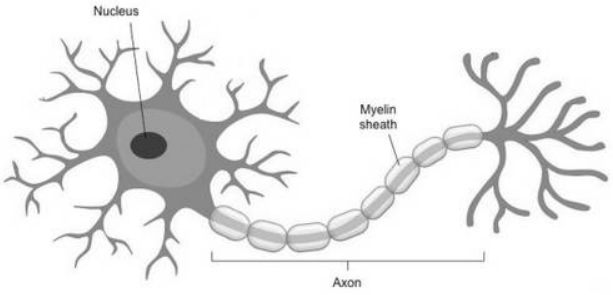
2024 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify the independent or dependent variable

Answer 8 of 10 · Question 3 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	

Type MS · Identify the independent or dependent variable

Answer 9 of 10 · Question 7 part b · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Identify the independent or dependent variable

Answer 9 of 10 · Question 7 part b · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Identify the independent or dependent variable

Answer 10 of 10 · Question 5 part a.i · 1 mark

2024 JUN · Paper 02 · find part a.i in this mark scheme extract

Number				
5	(a) (i)	**C** the type of antibiotic added The following are incorrect because: A the mass of bacteria added is irrelevant B the size of the inhibition zone is the dependent variable D the mass of agar in the Petri dish is irrelevant		(1)
	(ii)	Any two from: • sterilise equipment/named equipment/disinfect/swab desk; • use of sterile agar; • reference to use of lid on Petri dish/ minimising the time the lid is removed; • Flame from Bunsen burner:		(2)
	(iii)	(antibiotic) Q; greater inhibition zone /greater clear zone/ more bacteria killed;		(2)
	(iv)	Any four from: • mutation; • (resistant) bacteria survive; • reproduce; • pass on resistant gene/allele; • Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations)		(4)
	(b)	• (bacteria/fungi/decomposers) release enzymes/named enzyme • breakdown dead/organic matter/ named organic matter; • release nutrients/named nutrients into the soil/environment;	Ignore carbon dioxide Ignore fertiliser	(2)

Total for Question 5 = 11 marks